

Planck-99

TinyML Binary Malware Classifier for Embedded Linux & IoT — Technical Brief

96.28%	34 ns	27 KB	1.4 MB
Accuracy — unseen IoT	Median C-kernel latency	Model binary footprint	Peak RAM (C kernel)

01 — What Planck-99 Is

Planck-99 is a deterministic, integer-only binary classifier that ingests a host-level syscall trace and returns a single verdict: **Malware** or **Benign**. It is compiled to a standalone C header — no runtime, no interpreter, no OS calls — and runs entirely in userspace with zero floating-point operations and zero external dependencies. Every classification produces a JSON audit trail with the full decision path.

The design target is **first-line detection on resource-constrained embedded Linux and IoT devices** — hardware classes that rule out traditional AV, cloud pipelines, or GPU-accelerated inference. Planck-99 v1 is the initial release; the limitations section documents what future versions will improve.

02 — Architecture Decisions & Rationale

Why a decision-tree ensemble, not a neural network?

The deployment constraint is a C compiler and integer arithmetic — nothing else. Neural networks require matrix multiplication in floating-point or dedicated quantization runtimes (TFLite, ONNX RT) that add hundreds of KB of dependency. A pruned decision tree ensemble compiles to a nested if-else structure in ~4 KB of C — fully auditable, branch-predictable, and cache-resident on any modern MCU. Every inference path can be traced and verified statically.

Why 5 trees after pruning?

Training begins with 300 trees. Post-training pruning identifies the minimal sub-ensemble that preserves accuracy on the validation set within tolerance. The result is 5 trees — small enough to fit in L1 cache on most embedded cores, large enough to provide majority-vote stability (3-of-5 decision boundary). Adding more trees past this point yielded no measurable accuracy gain on held-out data, confirming the ensemble has reached the bias-variance optimum for this feature space.

Why a fixed 32-dimensional feature vector?

Raw syscall sequences are variable-length strings — unusable directly for fixed-size classifiers. The engineering decision is to project any sequence, regardless of length, onto a fixed 32-dimensional space of normalized frequency ratios and statistical aggregates. This projection is the core of what makes the model length-invariant (explained in Section 05). The 32-dimensional size is the result of iterative feature selection: it is the smallest dimensionality that retains the inter-class separability needed for the accuracy targets.

03 — Training Dataset

The model was trained on **4,600 real-world syscall samples** — **2,300 malware / 2,300 benign**, a perfectly balanced split. No synthetic data was used. Malware samples span 2016–2026, covering a decade of Linux threat evolution with no temporal overfitting observed at evaluation time.

Chopped fixed-window sequences — the critical design choice

Each training sample is not a complete execution trace. It is a **fixed-window sub-sequence chopped from a real trace**. This is deliberate: by training on bounded windows rather than full runs, the model is forced to learn frequency fingerprints that are statistically stable in any window of sufficient size. The consequence is that the trained model generalizes to traces far longer than anything it saw during training — including traces 51× the training ceiling — without any retraining.

Split	Samples	Class balance	Median length	Max length
Training + Validation	4,600	50% / 50% (exact)	863 syscalls	2,275 syscalls

04 — Quantization Trade-off: INT8 vs Float32

The Python training pipeline operates in float32. The deployed C kernel uses INT8 fixed-point throughout — no floating-point unit required at inference time. The quantization step introduces a controlled accuracy loss in exchange for the binary size and latency profile that makes embedded deployment feasible.

What quantization costs

Across all three evaluation datasets, the INT8 C kernel loses at most **0.5 percentage points** in accuracy compared to the float32 Python model. This is the entire cost of quantization — measured empirically on held-out data, not estimated.

What quantization buys

- Binary size: 27 KB compiled vs ~1 GB Python environment.
- Peak RAM: 1.4 MB vs 967 MB — a 690× reduction.
- Latency: 34–62 ns median vs 17–538 ms Python — up to 15.8 million× faster.
- FPU dependency: eliminated entirely. Runs on any core with integer ALU.
- Deployment model: single header file, any C compiler, zero runtime.

Metric	Float32 (Python)	INT8 (C kernel)	Delta
Accuracy — train validation	98.97%	98.88%	−0.09 pp
Accuracy — LinuxStatic	92.42%	92.01%	−0.41 pp
Accuracy — IoT (ADFA-LD)	96.48%	96.28%	−0.20 pp
Peak RAM	967 MB	1.4 MB	690× reduction
Inference latency (IoT)	538 ms	34 ns	15,800,000× faster
Binary / model size	~1 GB env	27 KB	—
FPU required	Yes	No	—

05 — Out-of-Distribution Generalisation: Why Length-Invariance Works

The model was trained on traces up to 2,275 syscalls. On ADFA-LD it achieves 97.87% recall on traces up to **117,088 syscalls — a 51× extrapolation** beyond the training ceiling, with no retraining or fine-tuning. This result requires explanation.

The mathematical intuition

The 32-dimensional feature vector is built entirely from **normalized frequency ratios and statistical aggregates**, not from raw counts or positional features. A normalized ratio — such as the proportion of file-I/O syscalls out of total syscalls, or the Shannon entropy of the call distribution — converges to a stable value as the sequence grows and remains statistically consistent across any window once a minimum sample size is reached.

Concretely: if a malware process makes file-I/O calls 54% of the time, that ratio is approximately 0.54 whether measured over 500 syscalls or 50,000. The feature vector captures *what kind of process this is*, not *how long it ran*. The decision boundary learned during training on short windows transfers intact to long traces because the underlying statistical signature is the same.

Why this also explains the short-trace failure mode

The same property that enables long-trace generalization also defines the failure regime. Frequency ratios are estimates from a sample. Below ~500 syscalls, the sample is too small for the ratio to stabilize — high-variance estimates push the feature vector away from its true population value and across the decision boundary. This is a statistical convergence issue, not a model capacity issue, and it is why the minimum-length gate is mandatory rather than optional.

Trace length range	Feature ratio stability	Error rate	Action
< 500 syscalls	Unstable — high variance	16%–30%	Buffer; do not classify
500–1,200 syscalls	Converging	5%–10%	Classify with caution
1,200–2,275 syscalls	Stable (training regime)	< 2%	Optimal window
> 2,275 syscalls	Stable (OOD, confirmed)	< 2%	Generalises well

06 — Benchmark Results

Three datasets. The internal set validates in-distribution performance. LinuxStatic ELF and IoT Syscalls (ADFA-LD) are fully external — zero overlap with training data. All results are from the C kernel (INT8) unless noted.

Dataset	Type	Accuracy	Precision	Recall	F1	FPR
Training validation	Internal	98.88%	98.58%	99.21%	98.90%	1.45%
LinuxStatic ELF	External — unseen	92.01%	100.00%	92.01%	95.84%	0.00%
IoT Syscalls (ADFA-LD)	External — unseen	96.28%	97.71%	97.87%	97.79%	12.18% *

ADFA-LD dataset profile

ADFA-LD is a public Linux IDS benchmark (UNSW). The evaluated subset contains 1,505 samples with a heavily skewed 1:5.3 benign-to-malware ratio — only 238 benign traces against 1,267 malware traces. Benign traces have a median of 2,234 syscalls; malware extends to 117,088 (51× training ceiling). The dataset is used here as a stress-test for OOD generalization, not as a balanced benchmark.

Latency (C kernel — 9 independent runs, $\sigma \leq 14$ ns)

Dataset	Median trace length	C kernel median	Python baseline	Speedup
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Training validation	863 syscalls	62 ns	17 ms	≈274,000×
LinuxStatic ELF	1,161 syscalls	35 ns	18 ms	≈510,000×
IoT Syscalls (ADFA-LD)	15,075 syscalls	34 ns	538 ms	≈15,800,000×

Latency is trace-length invariant by design — the C kernel receives a fixed 32-feature vector regardless of original trace length.

07 — Known Limitations

L1 — Full sequence required per classification pass

Planck-99 v1 classifies a complete buffered window, not an incrementally updated stream. A process must accumulate at least 500 syscalls before any verdict is possible, and the recommended window is 1,200+ syscalls for reliable operation. The deployment model is: *collect* → *gate* → *classify* → *slide window by 50%* → *repeat*. This means early-lifecycle processes cannot be classified until sufficient evidence accumulates. Future versions will address this with an online feature-update architecture.

L2 — 12.18% false positive rate on IoT benign traces

On ADFA-LD the model raises 29 false alarms from 238 benign samples (12.18% FPR). The root cause is a distribution mismatch, not a model flaw:

- The 238 benign traces have a median of only 2,234 syscalls — sitting at the lower edge of the stable frequency-ratio window. The frequency fingerprint has not fully converged, pushing some benign samples toward the malware side of the decision boundary.
- The benign traces are dominated by generic system calls (read, mmap, fstat) whose frequency profiles overlap with certain malware behavioral signatures at short lengths.
- The 1:5.3 class imbalance in ADFA-LD means benign is severely underrepresented; on the balanced internal dataset FPR is 1.45%.

Accepted trade-off rationale: in the target deployment context, a missed malware detection carries significantly higher operational cost than a false alarm. A false positive triggers a review event; an undetected intrusion on an air-gapped embedded device may be unrecoverable. At this model size, latency, and accuracy profile, the v1 FPR is an accepted trade-off for a first-line detector. The 500-syscall gate eliminates most of this artifact in production deployments where traces are continuously buffered.

Future improvement path: expand benign training data with short IoT-representative traces to close the distribution gap and reduce FPR toward the 1.45% baseline.

L3 — No incremental / streaming classification in v1

The current architecture requires a complete window for each inference pass. Streaming updates to the feature vector as syscalls arrive would allow earlier detection and eliminate the buffering latency. This is a planned v2 capability.

08 — Deployment Model

The intended deployment is a lightweight controller loop running alongside the kernel syscall logger on any embedded Linux SoC. Total system footprint is approximately **37 KB** (10 KB ring buffer + 27 KB C kernel binary) — fits in the SRAM of any modern embedded Linux device without a dedicated AI accelerator.

Step	Component	Detail
1	Syscall logger	Kernel hook appends each syscall name to a ring buffer

2	Length gate	Buffer < 500 syscalls → keep buffering, no classification
3	Feature extraction	32-dimensional normalized vector computed from buffer
4	C kernel inference	34–62 ns; integer-only, zero FPU, zero network
5a	On Malware verdict	Raise alert + write JSON audit trail
5b	On Benign verdict	Slide window by 600 syscalls (50% overlap); go to step 2

09 — Competitive Position

Capability	Traditional AV	Cloud detection	Planck-99 v1
Peak RAM	50–500 MB	Server-side	1.4 MB
Inference latency	10–100 ms	200–2,000 ms	34 ns
Network required	No (offline sig)	Mandatory	Never
FPU / GPU required	No / No	Yes / Yes	Never
MCU-class deployable	No	No	Yes
Air-gap safe	No	No	Yes
Per-decision audit	No	No	JSON proof trail
Accuracy on unseen IoT	95–98% (est.)	N/A	96.28%
False positive rate	Vendor-dependent	N/A	1.45% (balanced)

10 — Roadmap

Priority	Improvement	Target outcome
High	Online / streaming feature updates	Verdicts without fixed-window buffering
High	Expanded benign training data (IoT)	FPR < 3% on short benign traces
Medium	ARM / RISC-V embedded benchmarks	Validated latency on MCU-class hardware
Medium	Multi-class detection (v2)	Malware family identification
Low	Relaxed minimum sequence gate	Reliable inference below 500 syscalls